



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2018 HSC Course Assessment Task 4 (Trial HSC)

Thursday, 9 August 2018

General instructions

- Working time – 2 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt all questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.
- A NESA Reference Sheet is provided.

SECTION I

- Mark your answers on the answer grid provided

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT NUMBER: **# BOOKLETS USED:**

Class: (please ✓)

☐ 12M1 – Mr Sekaran

☐ 12M2 – Mrs Bhamra

☐ 12M3 – Mr Tan

☐ 12M4 – Mrs Gan

☐ 12M5 – Mr Lam

Marker's use only

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$	

Section I

10 marks

Attempt Question 1 to 10

Allow about 15 minutes for this section

Mark your answers on the multiple-choice answer grid

1. The point P divides the interval from $A(-3, 2)$ to $B(4, -7)$ externally in the ratio 5:3. 1

What is the x -coordinate of P ?

- (A) $14\frac{1}{2}$ (B) 13 (C) $1\frac{3}{8}$ (D) $-13\frac{1}{2}$

2. Which expression is equal to $\cos(A-B)\cos(A+B) - \sin(A-B)\sin(A+B)$? 1

- (A) $\cos(-2B)$ (B) $\cos 2A$
(C) $2\cos A$ (D) $\pi - \cos 2B$

3. What are the asymptotes of $y = \frac{2x^2}{1-x^2}$? 1

- (A) $x = \pm 1$ (B) $x = 2, y = \pm 1$
(C) $y = -2, x = \pm 1$ (D) $y = 2, x = \pm 1$

4. What is the value of $\lim_{x \rightarrow 0} \frac{\sin \frac{3x}{2}}{6x}$? 1

- (A) $\frac{1}{4}$ (B) 4 (C) $\frac{1}{2}$ (D) 2

5. What is the derivative of $\log_e \left(\frac{2x+1}{3x+2} \right)$? 1

- (A) $\frac{2}{3}$ (B) $\log_e \left(\frac{2}{3} \right)$ (C) $\frac{1}{(2x+1)(3x+2)}$ (D) $\frac{-1}{(2x+1)(3x+2)}$

6. The polynomial $P(x) = x^3 + ax^2 + 2x + 1$ has a factor $(x + a)$. 1

What is the value of a ?

- (A) $-\frac{1}{2}$ (B) ± 1 (C) $\frac{1}{2}$ (D) $\pm \frac{1}{\sqrt{2}}$

7. What is the general solution of $(\sin 2x - 1)(\cos 2x - 2) = 0$? 1

- (A) $x = n\pi + (-1)^n \frac{\pi}{4}$, n is an integer (B) $x = \frac{n\pi}{2} \pm \frac{3\pi}{4}$, n is an integer
(C) $x = n\pi + (-1)^n \frac{\pi}{2}$, n is an integer (D) $x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$, n is an integer

8. What is the inverse function of $f(x) = e^{x^3}$? 1

- (A) $f^{-1}(x) = 3e^x$ (B) $f^{-1}(x) = 3\log_e x$
(C) $f^{-1}(x) = \sqrt[3]{\log_e x}$ (D) $f^{-1}(x) = \sqrt{3\log_e x}$

9. A particle is moving in simple harmonic motion with displacement x . Its velocity v is given by $v^2 = 9(4 - x^2)$. What is the amplitude, A and the period, T of the motion? 1

- (A) $A = 3$ and $T = \pi$ (B) $A = 3$ and $T = \frac{\pi}{2}$
(C) $A = 2$ and $T = \frac{\pi}{3}$ (D) $A = 2$ and $T = \frac{2\pi}{3}$

10. What is the value of k such that $\int_2^k \frac{dx}{\sqrt{16 - x^2}} = \frac{\pi}{6}$? 1

- (A) $\frac{2\pi}{3}$ (B) $2\sqrt{3}$ (C) $\frac{4\pi}{3}$ (D) $\frac{\sqrt{3}}{2}$

Section II

60 marks

Attempt Questions 11 to 14

Allow about 1 hour and 45 minutes for this section

Write your answers in the writing booklets supplied. Additional writing booklets are available.
Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence on a NEW booklet	Marks
(a)	Evaluate $\sin^{-1}\left(\cos\frac{4\pi}{3}\right)$.	2
(b)	The polynomial equation $x^3 - 3x^2 - 2x + 5 = 0$ has roots α , β and γ . Find the value of $\alpha^2 + \beta^2 + \gamma^2$.	2
(c)	Differentiate $x^2 \sin^{-1} 2x$	2
(d)	Using the expansion of $\sin(\theta + \phi)$ prove that $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$	2
(e)	Solve the inequality $\frac{3}{2x-4} \geq -2$	3
(f)	(i) Show that $\frac{1}{e^x + 4e^{-x}} = \frac{e^x}{e^{2x} + 4}$.	1
	(ii) Hence find $\int \frac{4}{e^x + 4e^{-x}} dx$ by using the substitution $u = e^x$.	3

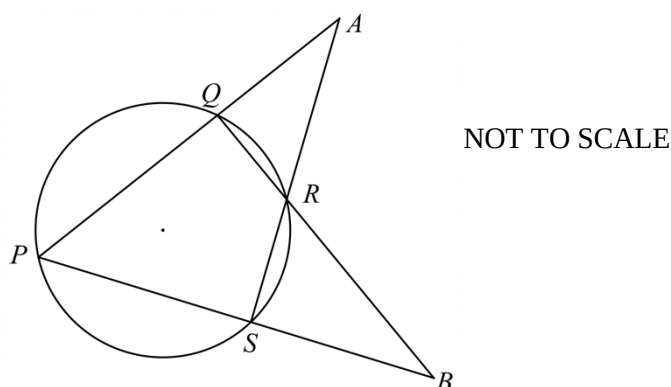
Question 12 (15 Marks) Commence on a NEW booklet**Marks**

- (a) One of the roots of the equation $x^3 + kx^2 + 1 = 0$ is the sum of the other two roots.

(i) Show that $x = -\frac{k}{2}$ is a root of the equation 2

(ii) Find the value of k . 2

- (b) In the diagram, PQA , PSB , BRQ and SRA are straight lines and $\angle PAS = \angle PBQ$

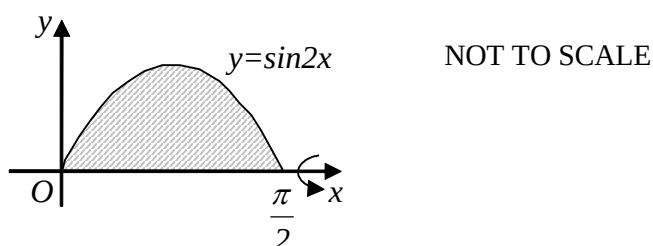


Copy or trace the diagram into your writing booklet.

(i) Prove that $\angle PQR = \angle PSR$ 2

(ii) Hence, or otherwise, prove that PR is a diameter. 2

- (c) The region bounded by $y = \sin 2x$ and the x -axis, between $x = 0$ and $x = \frac{\pi}{2}$, is rotated about the x -axis to form a solid. 3



Find the volume of the solid.

- (d) (i) Starting with the graph of $y = e^{2x}$ show by a sketch that the equation $e^{2x} + 4x - 5 = 0$ has only one solution. 2

(ii) Taking $x_1 = 0.5$ as a first approximation of the root of the equation $e^{2x} + 4x - 5 = 0$, use one step of Newton's method to find a better approximation correct to 2 decimal places. 2

Question 13 (15 Marks) Commence on a NEW booklet**Marks**

- (a) A sphere is being heated so that its surface area is increasing at a rate of $1.5 \text{ mm}^2 \text{ s}^{-1}$.

Using the formulae for surface area and volume as $A = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$ respectively:

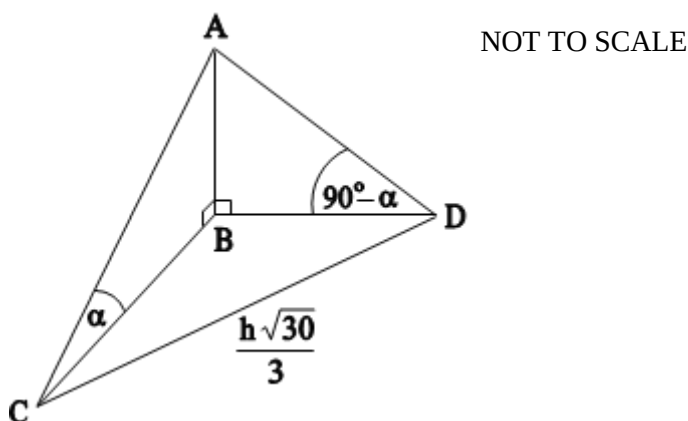
(i) Show that $\frac{dr}{dt} = \frac{3}{16\pi r}$. 2

(ii) Hence find the rate at which the volume is increasing when the radius is 60 mm . 2

- (b) Charles is at point C south of a tower AB of height h metres. His friend Daniel is at a point D, which is closer to the tower and east of it.

The angles of elevation of the top A of the tower from Charles and Daniel's positions are α and $90^\circ - \alpha$ respectively.

The distance CD between Charles and Daniel is $\frac{h\sqrt{30}}{3}$ metres.



(i) Show that $3\tan^4\alpha - 10\tan^2\alpha + 3 = 0$ 3

(ii) Hence, find α , the angle of elevation at which Charles can see the top A of the tower. 2

- (c) Use mathematical induction to prove that for all integers $n \geq 1$ 3

$$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$

Question 13 continued overleaf...

- (d) A particle is moving along the x -axis. Its displacement x at time t is given by

$$x = 2 - 2 \sin\left(3t + \frac{\pi}{3}\right).$$

- (i) Find an expression for acceleration of the particle in terms of x . 2
- (ii) Hence explain why the motion of this particle is simple harmonic. 1

Question 14 (15 Marks) Commence on a NEW booklet

Marks

- (a) The rate at which a substance evaporates is proportional to the amount of the substance which has not yet evaporated. That is $\frac{ds}{dt} = -k(s - A)$; where A is the initial amount of substance, s is the amount which has evaporated at time t and k is a constant.

- (i) Show that $s = A(1 - e^{-kt})$ satisfies the equation $\frac{ds}{dt} = -k(s - A)$ 1

- (ii) Sketch the graph s against t . 2

- (iii) Show that the time it takes for $\frac{7}{8}$ of the substance to evaporate is $\frac{3}{k} \ln 2$. 2

- (b) The acceleration of a particle P is given by $\ddot{x} = 4x(x^2 + 2)$, where x metres is the displacement of P from a fixed point after t seconds. Initially, the particle is at O and has velocity $v = 2\sqrt{2} \text{ ms}^{-1}$.

- (i) Show that $v^2 = 2(x^2 + 2)^2$. 3

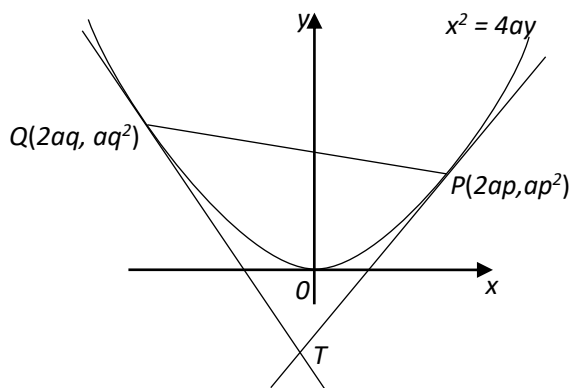
- (ii) Explain why the particle will always travel in a positive direction. 1

- (iii) By finding an expression for $\frac{dt}{dx}$, or otherwise, find x as a function of t . 2

Question 14 continued overleaf...

- (c) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

The tangents to the parabola at P and Q intersect at the point T . The coordinates of the point T is given by $x = a(p + q)$ and $y = apq$. (Do Not prove this.)



NOT TO SCALE

- | | |
|---|---|
| (i) Show that $p - q = 1 + pq$ if the tangents at P and Q intersect at 45° | 2 |
| (ii) Find the Cartesian equation of the locus of T | 2 |

End of paper

Normanhurst Boys High School
2018 Higher School Certificate
Trial Examination

Mathematics Extension1
Marking guidelines

Section I

Question	Sample Answer	Marking Key
1	$x = \frac{mx_2 + nx_1}{m+n}$ $= \frac{3(-3) + -5(4)}{3-5}$ $= \frac{-29}{-2}$ $= 14\frac{1}{2}$	A
2	$\cos[(A-B) + (A+B)]$ $= \cos 2A$	B
3	$1-x^2=0$ $x=\pm 1$ $\lim_{x \rightarrow \infty} \frac{2x^2}{1-x^2}$ $= \lim_{x \rightarrow \infty} \frac{2}{\frac{1}{x^2}-1}$ $= \frac{2}{-1}$ $= -2$	C
4	$\lim_{x \rightarrow 0} \frac{\sin \frac{3x}{2}}{4 \cdot \frac{6x}{4}} = \frac{1}{4} \lim_{x \rightarrow 0} \frac{\sin \frac{3x}{2}}{\frac{3x}{2}}$ $= \frac{1}{4}$	A
5	$\frac{d}{dx} [\log_e(2x+1) - \log_e(3x+2)]$ $= \frac{2}{2x+1} - \frac{3}{3x+2}$ $= \frac{2(3x+2) - 3(2x+1)}{(2x+1)(3x+2)}$ $= \frac{1}{(2x+1)(3x+2)}$	C

6	$P(-a) = 0, \quad (-a)^3 + a(-a)^2 + 2(-a) + 1 = 0$ $-a^3 + a^3 - 2a + 1 = 0$ $1 = 2a$ $\therefore a = \frac{1}{2}$	C
7	$\sin 2x = 1, \quad \cos 2x - 2 \neq 0$ $2x = n\pi + (-1)^n \frac{\pi}{2}$ $x = \frac{n}{2}\pi + (-1)^n \frac{\pi}{4}$	D
8	$x = e^{y^3}$ $\log_e x = y^3$ $\therefore y = \sqrt[3]{\log_e x}$	C
9	$v^2 = 3^2 (z^2 - x^2)$ $n = 3, \quad a = 2$ $T = \frac{2\pi}{3}, \quad A = 2$	D
10	$\left[\sin^{-1}\left(\frac{x}{4}\right) \right]_2^K = \frac{\pi}{6}$ $\sin^{-1}\left(\frac{K}{4}\right) - \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$ $\sin^{-1}\left(\frac{K}{4}\right) - \frac{\pi}{6} = \frac{\pi}{6}$ $\sin^{-1}\left(\frac{K}{4}\right) = \frac{\pi}{3}$ $\frac{K}{4} = \sin\left(\frac{\pi}{3}\right)$ $K = 4\left(\frac{\sqrt{3}}{2}\right)$ $= 2\sqrt{3}$	B

Section II

Question 11 (a)

Criteria	Marks
• Provides correct solution	2
• Obtains $\sin^{-1}\left(-\frac{1}{2}\right)$	1

Sample answer

$$\begin{aligned}\sin^{-1}\left(\cos\frac{4\pi}{3}\right) &= \sin^{-1}\left(-\frac{1}{2}\right) \\ &= -\sin^{-1}\left(\frac{1}{2}\right) \\ &= -\frac{\pi}{6}\end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Obtains $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$	1

Sample answer

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= (3)^2 - 2(-2) \\ &= 13\end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Apply product rule correctly	1

Sample answer

$$2x \sin^{-1} 2x + \frac{2x^2}{\sqrt{1-4x^2}}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Correctly expanded $\sin(30^\circ + 45^\circ)$	1

Sample answer

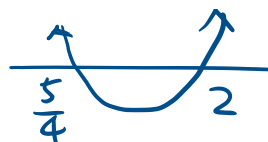
$$\begin{aligned}\sin(30^\circ + 45^\circ) &= \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ \\ \therefore \sin(75^\circ) &= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} \\ &= \frac{1+\sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}+\sqrt{6}}{4}\end{aligned}$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Achieves $(2x - 4)(4x - 5) \geq 0$	2
• Multiply both sides by $(2x - 4)^2$	1

Sample answer

$$\begin{aligned}
 (2x-4)^2 \cdot \frac{3}{2x-4} &\geq -2(2x-4)^2, \quad x \neq 2 \\
 3(2x-4) &\geq -2(2x-4)^2 \\
 (2x-4)[3 + 2(2x-4)] &\geq 0 \\
 2(x-2)(4x-5) &\geq 0 \\
 x \leq \frac{5}{4}, \quad x > 2
 \end{aligned}$$



Question 11 (f) (i)

Criteria	Mark
• Provides correct solution	1

Sample answer

$$\begin{aligned}
 \frac{1}{e^x + 4e^{-x}} &= \frac{1}{e^x + \frac{4}{e^x}} \\
 &= \frac{1}{\frac{e^{2x} + 4}{e^x}} \\
 &= \frac{e^x}{e^{2x} + 4}
 \end{aligned}$$

Question 11 (f) (ii)

Criteria	Marks
• Provides correct solution	3
• Achieves $\int \frac{4}{u^2 + 4} du$	2
• Achieves $du = e^x dx$	1

Sample answer

$$\begin{aligned}
 u &= e^x \\
 du &= e^x dx \\
 \int \frac{4}{e^x + 4e^{-x}} dx &= \int \frac{4e^x}{e^{2x} + 4} dx \\
 &= \int \frac{4}{u^2 + 4} du \\
 &= 4 \cdot \frac{1}{2} \tan^{-1}\left(\frac{u}{2}\right) + C \\
 &= 2 \tan^{-1}\left(\frac{e^x}{2}\right) + C
 \end{aligned}$$

Question 12 (a) (i)

Criteria	Mark
• Provides correct solution	2
• Achieves $\alpha + \beta + (\alpha + \beta) = -k$	1

Sample answer

let roots be α, β and $\alpha + \beta$
 sum of roots $\alpha + \beta + \alpha + \beta = -k$ ✓
 $\alpha + \beta = -\frac{k}{2}$ is a root ✓

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Substitute $x = -k/2$ into equation	1

Sample answer

$(-\frac{k}{2})^3 + k(-\frac{k}{2})^2 + 1 = 0$ ✓
 $-\frac{k^3}{8} + \frac{k^3}{4} + 1 = 0$
 $-k^3 + 2k^3 + 8 = 0$
 $k^3 = -8$
 $k = -2$ ✓

Question 12 (b) (i)

Criteria	Mark
• Provides correct solution	2
• Shows exterior angle $\angle PQR$ or $\angle PSR$ in terms of sum of interior angles or prove that $\triangle APS$ similar to $\triangle BPQ$	1

Sample answer

$\angle PAS = \angle PBQ$ (given) — ①
 $\angle ARQ = \angle BRS$ (vertically opposite angles) — ②
 $\angle PQR = \angle PAS + \angle ARQ$ (exterior angle of a triangle)
 $\angle PSR = \angle PBQ + \angle BRS$ (" " " ") } ✓
 ① + ② $\angle PAS + \angle ARQ = \angle PBQ + \angle BRS$
 $\therefore \angle PQR = \angle PSR$ ✓

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Shows $\angle PQR + \angle PSR = 180^\circ$ with reasons	1

Sample answer

$\angle PQR + \angle PSR = 180^\circ$ (opposite angles of cyclic quad.) ✓
 From (i) $\angle PQR + \angle PSR = 180^\circ$
 $\angle PQR = 90^\circ$ ✓
 $\therefore PR$ is a diameter (angle in a semicircle is 90°)

Question 12 (c)

Criteria	Marks
• Provides correct solution	3
• Correct integration	2
• Correct integral for volume in $\cos 4x$	1

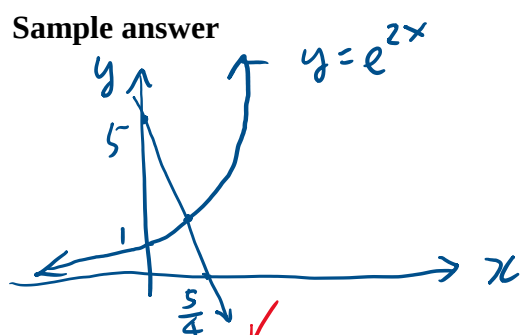
Sample answer

$$\begin{aligned}
 V &= \pi \int_0^{\frac{\pi}{2}} \sin^2 2x \, dx \\
 &= \pi \int_0^{\frac{\pi}{2}} \frac{1 - \cos 4x}{2} \, dx \quad \checkmark \\
 &= \frac{\pi}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{2}} \quad \checkmark \\
 &= \frac{\pi}{2} \left[\left(\frac{\pi}{2} - \frac{\sin 2\pi}{4} \right) - 0 \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{2} - 0 \right] \\
 &= \frac{\pi^2}{4} \quad \checkmark
 \end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct solution or correct sketch of $y = e^{2x} + 4x - 5$	2
• Correctly drawn two graphs	1

Sample answer



Graphs of $y = e^{2x}$ and $y = -4x + 5$ intersect only at one point ✓
 $\therefore e^{2x} = -4x + 5$
 ie $e^{2x} + 4x - 5 = 0$
 has only one solution

Question 12 (d) (ii)

Criteria	Mark
• Provides correct solution	2
• Correct substitution into formula	1

Sample answer

$$\begin{aligned}
 \text{let } f(x) &= e^{2x} + 4x - 5 \\
 f'(x) &= 2e^{2x} + 4 \\
 x_2 &= 0.5 - \frac{f(0.5)}{f'(0.5)} = 0.5 - \frac{e + 4(0.5) - 5}{2e + 4} \quad \checkmark \\
 &= 0.52985 \dots \\
 &\approx 0.53 \quad (2 \text{ d.p.}) \quad \checkmark
 \end{aligned}$$

Question 13 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Achieves $\frac{dA}{dt} = \frac{3}{2}$ and $\frac{dA}{dr} = 8\pi r$	1

Sample answer

$$\begin{aligned}
 \frac{dA}{dt} &= 1.5 \approx \frac{3}{2} \text{ mm s}^{-1} \\
 A &= 4\pi r^2 \rightarrow \frac{dA}{dr} = 8\pi r \\
 \frac{dr}{dt} &= \frac{dr}{dA} \times \frac{dA}{dt} \\
 &= \frac{1}{8\pi r} \times \frac{3}{2} \\
 &= \frac{3}{16\pi r}
 \end{aligned}$$

Question 13 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Achieves $\frac{dV}{dt} = 4\pi r^2 \times \frac{3}{16\pi r}$	1

Sample answer

$$\begin{aligned}
 V &= \frac{4}{3}\pi r^3 \\
 \frac{dV}{dr} &= 4\pi r^2 \\
 \frac{dr}{dt} &= \frac{dV}{dr} \times \frac{dr}{dt} \\
 &= 4\pi r^2 \times \frac{3}{16\pi r} \\
 &= \frac{3r}{4} \quad \checkmark \\
 \text{when } r &= 60 \text{ mm} \quad \frac{dV}{dt} = \frac{3(60)}{4} \\
 &= 45 \text{ mm}^3 \text{ s}^{-1} \quad \checkmark
 \end{aligned}$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Correct substitution into Pythagoras' equation	2
• Correctly find BC and BD	1

Sample answer

$$\begin{aligned} \triangle ABC, \tan \alpha &= \frac{h}{BC} \rightarrow BC = \frac{h}{\tan \alpha} \\ \triangle ABD, \tan(90 - \alpha) &= \frac{h}{BD} \rightarrow BD = \frac{h}{\tan(90 - \alpha)} \\ &= \frac{h}{\cot \alpha} \\ &= h \tan \alpha \quad \checkmark \\ \angle CBD &= 90^\circ \\ BC^2 + BD^2 &= CD^2 \\ \frac{h^2}{\tan^2 \alpha} + h^2 \tan^2 \alpha &= h^2 \left(\frac{30}{9} \right) \quad \checkmark \\ \frac{1}{\tan^2 \alpha} + \tan^2 \alpha - \frac{10}{3} &= 0 \\ \times 3 \tan^2 \alpha \quad 3 + 3 \tan^4 \alpha - 10 \tan^2 \alpha &= 0 \\ \text{i.e. } 3 \tan^4 \alpha - 10 \tan^2 \alpha + 3 &= 0 \quad \checkmark \end{aligned}$$

Question 13 (b) (ii)

Criteria	Mark
• Provides correct solution	2
• Correctly solved for two values of $\tan \alpha$	1

Sample answer

$$\begin{aligned} (3 \tan^2 \alpha - 1)(\tan^2 \alpha - 3) &= 0 \\ \tan^2 \alpha &= \frac{1}{3} \text{ or } 3 \\ \tan \alpha &= \frac{1}{\sqrt{3}} \text{ or } \sqrt{3} \text{ only as } \alpha \text{ is acute} \\ \alpha &= 30^\circ \text{ or } 60^\circ \quad \checkmark \end{aligned}$$

But Daniel is closer than Charles to point B

$$\begin{aligned} 90 - \alpha &> \alpha \\ 90 &> 2\alpha \\ \text{i.e. } \alpha &< 45^\circ \\ \therefore \alpha &= 30^\circ \quad \checkmark \end{aligned}$$

Question 13 (c)

Criteria	Marks
• Provides correct proof	3
• Establish induction step, or equivalent merit	2
• Establish initial case, or equivalent merit	1

Sample answer

$$\text{Let } T_n = \frac{1}{(3n-2)(3n+1)}, S_n = \frac{n}{3n+1}$$

$$\text{When } n=1, U_1 = \frac{1}{1 \times 4} = \frac{1}{4}$$

$$RHS = \frac{1}{3+1} = \frac{1}{4} \therefore \text{True for } n=1$$

Assume true for some integer k i.e. $S_k = \frac{k}{3k+1}$

$$\text{RTP } S_{k+1} = \frac{k+1}{3(k+1)+1} \text{ i.e. } \frac{k+1}{3k+4}$$

$$\begin{aligned} \text{Now } S_{k+1} &= S_k + T_{k+1} \\ &= \frac{k}{3k+1} + \frac{1}{(3(k+1)-2)(3(k+1)+1)} \\ &= \frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)} \\ &= \frac{k(3k+4) + 1}{(3k+1)(3k+4)} \\ &= \frac{3k^2 + 4k + 1}{(3k+1)(3k+4)} \\ &= \frac{(3k+1)(k+1)}{(3k+1)(3k+4)} \\ &= \frac{k+1}{3k+4} \end{aligned}$$

$\therefore$ True for $n=k+1$

Hence the result is true for all positive integers n by induction.

complete proof ✓

Question 13 (d) (i)

Criteria	Mark
• Provides correct solution	2
• Correct acceleration in terms of t	1

Sample answer

$$\begin{aligned}
 x &= 2 - 2 \sin\left(3t + \frac{\pi}{3}\right) \quad \text{--- ①} \\
 \dot{x} &= -6 \cos\left(3t + \frac{\pi}{3}\right) \\
 \ddot{x} &= 18 \sin\left(3t + \frac{\pi}{3}\right) \quad \checkmark \\
 &= 9 \left(2 \sin\left(3t + \frac{\pi}{3}\right)\right) \\
 &= 9(2 - x) \quad \text{from ①} \quad \checkmark
 \end{aligned}$$

Question 13 (d) (ii)

Criteria	Mark
• Provides correct proof	1

Sample answer

$$\begin{aligned}
 \ddot{x} &= -3^2(x-2) \\
 \therefore \text{particle moves in SHM about } x=2
 \end{aligned}$$

Question 14 (a) (i)

Criteria	Mark
• Provides correct proof	1

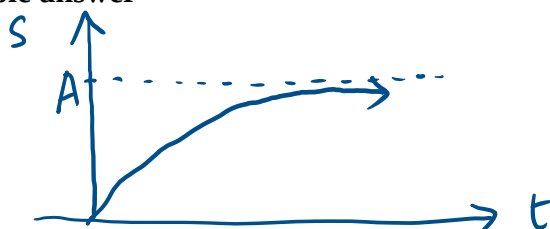
Sample answer

$$\begin{aligned}
 S &= A(1 - e^{-kt}) \rightarrow S - A = -Ae^{-kt} \\
 \frac{dS}{dt} &= Ak e^{-kt} \quad - (S - A) = Ae^{-kt} \\
 &= -k(S - A)
 \end{aligned}$$

Question 14 (a) (i)

Criteria	Marks
• Provides correct graph	2
• Notes asymptote at $s=A$ & labelled	1

Sample answer



Question 14 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Achieves $e^{-kt} = \frac{1}{8}$	1

Sample answer

$$\begin{aligned}
 \frac{7}{8}A &= A(1 - e^{-kt}) \\
 e^{-kt} &= 1 - \frac{7}{8} \\
 &= \frac{1}{8} \quad \checkmark \\
 -kt &= \ln\left(\frac{1}{8}\right) \\
 &= -\ln(2^3) \\
 &= -3\ln 2 \\
 t &= \frac{3\ln 2}{k} \quad \checkmark
 \end{aligned}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Achieves $\frac{1}{2}v^2 = (x^2 + 2)^2 + c$	2
• Achieves $\frac{1}{2}v^2 = \int 4x(x^2 + 2)dx$	1

Sample answer

$$\begin{aligned}
 \ddot{x} &= \frac{d}{dx}\left(\frac{1}{2}v^2\right) = 4x(x^2 + 2) \quad \checkmark \\
 \frac{1}{2}v^2 &= \int 4x^3 + 8x \, dx \\
 &= x^4 + 4x^2 + c \quad \checkmark \\
 \text{At } x=0, v &= 2\sqrt{2} \\
 \frac{1}{2}(2\sqrt{2})^2 &= 0 + c \\
 \therefore c &= 4 \\
 \frac{1}{2}v^2 &= x^4 + 4x^2 + 4 \\
 &= (x^2 + 2)^2 \\
 v^2 &= 2(x^2 + 2)^2 \quad \checkmark
 \end{aligned}$$

Question 14 (b) (ii)

Criteria	Mark
• Provides correct reasoning	1

Sample answer

$$v^2 = 2(x^2 + 2)^2$$

$$v = \pm \sqrt{2}(x^2 + 2)$$

as $v = 2\sqrt{2}$ when $x = 0$

$$v = \sqrt{2}(x^2 + 2) > 0 \quad \forall x$$

$\therefore$ the particle will always travel in a positive direction

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Achieves $t = \int \frac{dx}{\sqrt{2}(x^2 + 2)}$	1

Sample answer

$$v = \pm \sqrt{2}(x^2 + 2)$$

$$x = 0, v = 2\sqrt{2}$$

$$\therefore v = \sqrt{2}(x^2 + 2)$$

$$\frac{dx}{dt} = \sqrt{2}(x^2 + 2)$$

$$\frac{dt}{dx} = \frac{1}{\sqrt{2}(x^2 + 2)}$$

$$t = \frac{1}{\sqrt{2}} \int \frac{dx}{x^2 + 2} \quad \checkmark$$

$$= \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) \right] + C$$

$$= \frac{1}{2} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C$$

when $t = 0, x = 0 \therefore C = 0$

$$t = \frac{1}{2} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right)$$

$$\tan(2t) = \frac{x}{\sqrt{2}}$$

$$x = \sqrt{2} \tan(2t) \quad \checkmark$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Correctly substitute p and q into equation $\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$	1

Sample answer

gradient of tangent at P, $m_1 = p$
 gradient of tangent at Q, $m_2 = q$

$$\frac{p - q}{1 + pq} = \tan 45^\circ \quad \checkmark$$

$$1 + pq = 1$$

$$p - q = 1 + pq \quad \checkmark$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Achieves equation in $p+q$ and pq	1

Sample answer

From (i) $(p - q)^2 = (1 + pq)^2$

$$p^2 + q^2 - 2pq = 1 + (pq)^2 + 2pq$$

$$p^2 + q^2 + 2pq = 1 + (pq)^2 + 6pq$$

$$(p + q)^2 = 1 + (pq)^2 + 6pq \quad \checkmark$$

$$\left(\frac{x}{a}\right)^2 = 1 + \left(\frac{y}{a}\right)^2 + 6\left(\frac{y}{a}\right)$$

$$\times a^2 \quad x^2 = a^2 + y^2 + 6ay$$

$$x^2 - y^2 = a^2 + 6ay \quad \checkmark$$